Europäisches Patentamt
European Patent Office
Office européen des brevets

European Patent Office
80298 MUNICH
GERMANY

Questions about this communication?
Contact Customer Services at www.epo.org/contact

M. Zardi & Co S.A.
Via G. B. Pioda, 6
6900 Lugano
SUISSE

Date 25.11.2021

Reference	Application No./Patent No.
EUR1620WEP	20794947.0 - 1001 / 3872532 PCT/KR2020000837
Applicant/Proprietor	
Ftlab Co., Ltd.	

Communication

The extended European search report is enclosed.

The extended European search report includes, pursuant to Rule 62 EPC, the supplementary European search report (Art. 153(7) EPC) and the European search opinion.

Copies of documents cited in the European search report are attached.

☒ 0 additional set(s) of copies of such documents is (are) enclosed as well.

Refund of the search fee
If applicable under Article 9 Rules relating to fees, a separate communication from the Receiving Section on the refund of the search fee will be sent later.

Should you wish to further prosecute this application in the examination phase, your attention is drawn to the provisions of Rule 70a EPC. An invitation to respond to the extended European search report will be issued shortly (R. 70(2) EPC).

Europäisches Patentamt • European Patent Office • Office européen des brevets

Europäisches Patentamt
European Patent Office
Office européen des brevets

SUPPLEMENTARY EUROPEAN SEARCH REPORT

Application Number
EP 20 79 4947

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X Y	WO 91/13371 A1 (RADON TESTING CORP OF AMERICA [US]) 5 September 1991 (1991-09-05) * figures 1, 2 * * page 12, line 2 - page 13, line 25 * * page 15, line 21 - page 16, line 20 * -----	1-9,17 10-12, 16,18-20	INV. G01T1/185 G01T1/178 G01T7/00
X A	US 4 853 536 A (DEMPSEY JOHN C [US] ET AL) 1 August 1989 (1989-08-01) * column 9, line 52 - column 10, line 22; figure 3 * -----	1 2-20	
Y	JP 2005 156463 A (MITSUBISHI HEAVY IND LTD) 16 June 2005 (2005-06-16) * figures 3, 4 * -----	10-12,16	
Y	KR 2017 0023601 A (FTLAB CO LTD [KR]) 6 March 2017 (2017-03-06) * paragraphs [0075] - [0087]; figures 3, 6 * -----	18-20	

TECHNICAL FIELDS SEARCHED (IPC)

G01T

The supplementary search report has been based on the last set of claims valid and available at the start of the search.

Place of search	Date of completion of the search	Examiner
The Hague	17 November 2021	Johnstone, John

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons

& : member of the same patent family, corresponding document

EPO FORM 1503 03.82 (P04C04)

ANNEX TO THE EUROPEAN SEARCH REPORT ON EUROPEAN PATENT APPLICATION NO. EP 20 79 4947

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report. The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

17-11-2021

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 9113371 A1	05-09-1991	AT 180066 T	15-05-1999
		AU 7497791 A	18-09-1991
		CA 2053237 A1	23-08-1991
		EP 0477308 A1	01-04-1992
		JP H04506867 A	26-11-1992
		US 5093570 A	03-03-1992
		WO 9113371 A1	05-09-1991
US 4853536 A	01-08-1989	CA 1307860 C	22-09-1992
		US 4853536 A	01-08-1989
JP 2005156463 A	16-06-2005	NONE	
KR 20170023601 A	06-03-2017	NONE	

Information on Search Strategy - Pilot phase (see OJ 2015, A86)
The type of information contained in this sheet may change during the pilot for improving the usefulness of this new service.

Application Number
EP 20 79 4947

TITLE: RADON SENSOR DEVICE USING POLYHEDRAL-SHAPED IONIZATION CHAMBER

APPLICANT: Ftlab Co., Ltd.

IPC CLASSIFICATION: G01T1/185, G01T1/178, G01T7/00

EXAMINER: Johnstone, John

CONSULTED DATABASES: INET, NPL

CLASSIFICATION SYMBOLS DEFINING EXTENT OF THE SEARCH:

IPC:

CPC: G01T1/178, G01T1/185

FI/F-TERMS:

KEYWORDS OR OTHER ELEMENTS FEATURING THE INVENTION:
Polyhedral radon ionisation chamber.

The examination is being carried out on the **following application documents**

Description, Pages

1-16 as published

Claims, Numbers

1-20 as published

Drawings, Sheets

1/8-8/8 as published

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 WO 91/13371 A1 (RADON TESTING CORP OF AMERICA [US]) 5 September 1991 (1991-09-05)

D2 US 4 853 536 A (DEMPSEY JOHN C [US] ET AL) 1 August 1989 (1989-08-01)

D3 JP 2005 156463 A (MITSUBISHI HEAVY IND LTD) 16 June 2005 (2005-06-16)

D4 KR 2017 0023601 A (FTLAB CO LTD [KR]) 6 March 2017 (2017-03-06)

1 Clarity (Art. 84 EPC).

1.1 The application does not meet the requirements of Article 84 EPC, because claim 16 are not clear.

1.2 Claim 16 depends on claim 1 and refers to a 'second porbe' feature. However, claim 1 does not include any such 'second probe' feature and their does not appear to be any antecedent basis in either of claims 1 or 16 for the introduction of such a feature.

2 Independent claim 1. Novelty (Art. 52(1) and 54 EPC).

2.1 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 is not new within the meaning of Article 54(1) and (2) EPC.

2.2 **D1** (see **figs. 1, 2**) discloses:

A radon sensor device using a polyhedral-shaped (**page 12, lines 2-10: "Although a circular electret is illustrated (Fig. 1), and described, it should be understood that in alternative embodiments in accordance with the invention, the electret can have another shape, for example, rectangular, square, etc. with the other mechanical elements correspondingly contoured to accommodate the selected electret shape"**) ionization chamber, the radon sensor device comprising:

- an ionization chamber having an open side and inner sides surrounded by a first conductor and generating an electrical field therein by applying bias power to the first conductor (**12; page 13, lines 1-2: "The body 12 and lid 18 are preferably made of electrically conductive material"**);

- a cover having a first side covered with a second conductor and closing the open side of the ionization chamber such that that first conductor disposed on the inner sides of the ionization chamber and the second conductor are electrically connected (**18; page 13, lines 1-2: "The body 12 and lid 18 are preferably made of electrically conductive material"**);

- a probe unit disposed in the ionization chamber and absorbing ion charges produced when alpha (α) decay occurs in the ionization chamber (**34; page 15, line 21 - page 16, line 20: "charged particles or ions are produce in the detector 10' by the collision of alpha particles with molecules of air... The electrically charged products are attracted to the electret"**); and

- a measurement circuit detecting an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit into a predetermined magnitude (**page 15, line 21 - page 16, line 20: "reading the change in the voltage on the electret, for example, by means of a voltmeter or an electrometer"**).

2.3 The subject-matter of claim 1 is therefore not new (Article 54(1) and (2) EPC).

2.4 The subject-matter of independent claim 1 is also disclosed by document **D2**/ **fig. 3; column 9, line 52 - column 10, line 22: "The chamber 10 can be made of any rigid material which is impermeable to radon, such as metal or plastic, but its inner surface must be electrically conductive to conduct away to ground any electrostatic charged caused by ions which attach to it during monitoring. The cover 30 can be made of any rigid material such**

as plastic or metal, but its inner surface must also be electrically conductive... The preferred shape of the electret 61, which is permanently electrostatically charged, is a disk, as shown".

3 **Dependent claims 2-20.**

3.1 Dependent claims 2-12 and 16-20 do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty and/or inventive step.

- The subject-matter of claim 2 is obvious from **D1/page 12, lines 2-10: "Although a circular electret is illustrated (Fig. 1), and described, it should be understood that in alternative embodiments in accordance with the invention, the electret can have another shape, for example, rectangular, square, etc. with the other mechanical elements correspondingly contoured to accommodate the selected electret shape"**. Wherein, claim 2 merely suggests a slight constructional change in the radon sensor of **D1** which comes within the scope of customary practice followed by persons skilled in the art, especially as the advantages thus achieved can be readily contemplated in advance. Moreover, regarding the particular shape of the radon sensor housing, the technique of appropriately choosing the shape, whether polyhedral or otherwise, of a detector for the intended problem is well-known in the art and known to the skilled person who would be able to readily conceive the advantages of the technique. As such, the particular shape forming the subject-matter of claim 2 is considered to be merely one of several straightforward possibilities which the skilled person would select, depending on the circumstances, without exercising inventive skill, in order to solve the problem posed.

- The subject-matter of claim 3 is known from **D1/page 13, lines 9-25: "electret canisters made of conductive/conductively coated material"; page 18, lines 4-8: "a conductive coating is applied to the surfaces within the detector"**.

- The subject-matter of claim 4 is obvious from **D1/page 13, lines 9-12: "The purpose of the conductive inside surface of the electret chamber is to form an electrical field between the charged surface of the electret and the conductive surface of the conductive housing"**.

- The subject-matter of claim 5 is known from **D1/page 12, line 22 - page 13, line 8: "the body 12 is pressed or threaded into the lid 18... Friction or screw threads between the adjacent surfaces holds the canister 10 in its assembled condition... Continuous conductive surfaces are preferred in the canister"**.

- The subject-matter of claim 6 is obvious from **D1/page 13, lines 1-8: "The body 12 and lid 18 are preferably made of electrically conductive material, including a conductive plastic, or a non-conductive material coated, at least in part so that a conductive surface is presented within the canister 10"**.

- The subject-matter of claim 7 is known from **D1/fig. 1, ref. 40**.

- The subject-matter of claim 8 is obvious from **D1**. Wherein, claim 8 merely suggests a slight constructional change in the radon sensor of **D1** which comes within the scope of customary practice followed by persons skilled in the art, especially as the advantages thus achieved can be readily contemplated in advance. Moreover, regarding the particular dimensions of the radon sensor and the size and number of air holes, the technique of appropriately choosing the dimensions of a sensor for the intended problem is well-known in the art and known to the skilled person who would be able to readily conceive the advantages of the technique. As such, the particular values forming the subject-matter of claim 8 are considered to be merely one of several straightforward possibilities which the skilled person would select, depending on the circumstances, without exercising inventive skill, in order to solve the problem posed.

- The subject-matter of claim 9 is known from **D1/page 15, line 21 - page 16, line 20: "charged particles or ions are produce in the detector 10' by the collision of alpha particles with molecules of air... The electrically charged products are attracted to the electret"**.

- The subject-matter of claims 10, 11, 12 and 16 is obvious from **D3/figs. 3, 4, refs. 1a, 1b**, in combination with **D1**.

- The subject-matter of claim 17 is known from **D1/page 13, lines 1-12: "The body 12 and lid 18 are preferably made of electrically conductive material, including a conductive plastic, or a non-conductive material coated, at least in part so that a conductive surface is presented within the canister 10... The purpose of the conductive inside surface of the electret chamber is to form an electrical field between the charged surface of the electret and the conductive surface of the conductive housing"; page 15, line 21 -**

page 16, line 20: "charged particles or ions are produce in the detector 10' by the collision of alpha particles with molecules of air... The electrically charged products are attracted to the electret... reading the change in the voltage on the electret, for example, by means of a voltmeter or an electrometer".

- The subject-matter of claims 18 and 19 is obvious from **D4/fig. 3, refs. 130, 140, 150a, 150b, 160; fig. 6; [0075-0087]**, in combination with **D1**.

- The subject-matter of claim 20 is obvious from **D4/fig. 6, ref. 140'; [0082: "The noise detector 140 'is electrically connected between the guard ring 130' and the ground, and maintains the potential of the guard ring 130 'slightly higher than the ground so that the ionization chamber 110' (DC) type leakage current generated between the first and second units 120 'to the ground side and detects AC type noise to be output to a second preamplifier 150b"; 0084: "The noise detector 140 'may be formed by electrically connecting at least any one of active elements such as a resistor, a capacitor, and a diode electrically connected in series, in parallel, or in a combination of series and parallel"; 0087: "when the differential amplification is performed through the differential amplifier 160 ', noise having the same phase introduced through the probe unit 120' and the noise detection unit 140 'can be effectively canceled, and high-The signal can be obtained accurately and quickly"]**, in combination with **D1**.

3.2 The subject-matter of dependent claims 13, 14 and 15 do not appear to be disclosed by any of the documents cited in the search report. If the features of at least one of these claims, in heir entirety, were to be added to a redrafted independent claim then a novel and inventive set of claims could be envisaged (i.e. claims 1 + 12 + [13 or 14 or (14+15)]).

3.2.1 Such a combination of features would not be disclosed by any combination of **D1-D4**.

3.2.2 However, if the applicant intends to use any of these features to overcome the objections outlined in this communication then they are reminded to indicate:

i) the technical effect of the feature;

ii) the problem it is designed to overcome; and

iii) reasons for its non-obviousness.

In doing so the applicant is also reminded to pay particular attention to the requirements of Articles 82 and 123(2) EPC.

3.2.3 It is noted that this suggestion is only for assisting the applicant in their decision on how to proceed. It in no way precludes consideration of alternative solutions submitted by the applicant. The responsibility for determining the text of the application (Article 113(2) EPC) and in particular for defining the subject-matter for which protection is sought remains with the applicant.

3.2.4 Finally, if the applicant decides to follow the suggestion given they are respectfully asked to amend the description accordingly to ensure that the text faithfully reflects the updated set of claims and that any further requirements indicated in sections 4 and 5 below are also included.

4 **Further requirements.**

4.1 The applicant should adapt the description to the claims on file and delete or amend any statements or expressions which throw doubt on the scope of protection (GL F-IV, 4.3). Failure to do so could lead to the refusal of the application under Article 97(2) EPC in combination with Article 84 EPC.

4.2 The applicant should amend the description as needed to fulfil the following requirements:

– The technical field of the invention (Rule 42(1)(a), GL F-II.4.2) must correspond to the amended set of claims.

– The summary of the invention (Rule 42(1)(c) and GL F-II.4.5) must correspond to the amended set of claims. If appropriate, statements such as “The invention is set out in the appended set of claims“ or "The invention is as defined in claim 1" can be applied instead of repeating the claims verbatim.

– All embodiments which are not covered by the claims anymore must be deleted, unless these embodiments can reasonably be considered to be useful for highlighting specific aspects of the amended subject-matter. In such a case, they must be prominently marked as not being part of the present invention (T 1808/06). For example, ways of carrying out the invention using features presented as alternatives to the features of the independent claim must either be deleted or the fact that they are not encompassed by the claims must be prominently stated. Furthermore, any combinations of features which do not fall under the scope of the amended independent claims must be prominently marked as not being part of the present invention. Statements such as “The following examples/aspects/embodiments X, Y, Z are not according to the invention and are present for illustration purposes only” may be used at the start of the description (or in the list of figures) when such a situation arises.

– All technical features of the independent claims shall not be presented as optional in the description. Terms such as “for example“, “may“, “can”, “exemplary“, “optionally“, “preferably“ and the like should be removed when preceding a feature of an independent claim.

– Claim-like clauses must be deleted prior to grant. The term “claim-like clauses“ means clauses present in the description which use claim language such as “according to the preceding clause“, “characterised in that“, … (GL F-IV, 4.4). These “claim-like clauses“ also include examples, embodiments or aspects presented as claim-like clauses. A simple conversion of these clauses into separate optional features will not be admitted, either.

5 Procedure

5.1 The applicant is invited to file new claims which take account of the above comments.

5.2 The independent claims should be in the two-part form in accordance with Rule 43(1) EPC, which in the present case would be appropriate, with those features known in combination from the prior art being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

Independent claim 1 should therefore be redrafted accordingly. If, however, the applicant is of the opinion that the two-part form would be inappropriate, reasons therefor should be provided. In addition, the applicant should ensure that it is clear from the description which features of the subject-matter of claim 1 are already known in combination from **D1-D4** (see EPC Guidelines F-IV, 2.3.2). Alternatively, if the requisite prior art is adequately identified and discussed in the description then the two-part form need not be insisted upon (see EPC Guidelines F-IV, 2.3.2).

5.3 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion (see Guidelines F-IV, 4.19).

5.4 To meet the requirements of Rule 42(1)(b) EPC, **D1-D4** should be identified in the description and the relevant background art disclosed therein should be briefly discussed.

5.5 In order to comply with the requirements of Rule 137(4) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines H III, 2.1).

5.6 Amendments should be made by filing replacement pages. Unnecessary recasting of the description should be avoided. An amended abstract is not required. The applicant should also take account of the requirements of Rule 50(1) EPC. According to Rule 50(1) in conjunction with Rule 49(8) EPC, amendments shall be typed or printed. Handwritten amendments may only be made in documents other than those replacing application documents (Rule 50(2) EPC); they may, for example, be used to fulfil the requirements of Rule 137(4) EPC (identifying amendments and indicating basis for them).

5.7 If the applicant so wishes, they can contact the examiner about this application by e-mail at the following address: jjohnstone@epo.org, with a view to preparing his reply (see Notice of the EPO of 12.09.00, OJ 10/2000, 458).

N.B. Information exchanged with the European Patent Office by e-mail has no legal force in proceedings under either the EPC or PCT, and thus cannot be validly used to perform any procedural act and in particular to comply with time limits in proceedings (see Guidelines A-VIII, 2.5 and Notice of the EPO of 12.09.00, OJ EPO 2000, 458). This means that e.g. requests and/or amended documents and/or observations under Art. 94(3) EPC must be filed before expiry of the time limit set by one of the official means of communication (e.g. by fax) pursuant to Rule 2(1) EPC (see also Special Edition N° 3, OJ EPO 2007, A3, A5 and OJ EPO 2009 182-188).